Batu's Practice ~~Questions~~ Answers - Math217

Standard Normal CDF

Table of cumulative probabilities $\Phi(z) = P(Z \leq z)$ for the standard normal distribution

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Question 1. IQ Scores

IQ scores are normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$.

(a) What proportion of people have an IQ above 130?

Let X denote the RV of a persons IQ $\rightarrow$ find $P(X>130)$

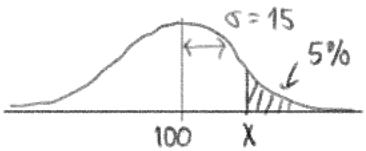
transform $\rightarrow z = \frac{130-\mu_X}{\sigma} = 2 \rightarrow P(X>130) = 1-P(X<130) = 1-\phi(2) \approx 0.0228$

(b) What proportion have an IQ between 85 and 115?

Want: $P(85<X<115) \rightarrow z_1 = \frac{85-\mu_X}{\sigma} = -1,\ z_2 = \frac{115-\mu_X}{\sigma} = 1$

$= \phi(1) - \phi(-1) = \phi(1) - (1-\phi(1)) = 2\phi(1) - 1 \approx 0.6826$

(c) What is the IQ score that separates the top 5% from the rest?



Want: z such that $\phi(z) = 1-0.05 = 0.95$ or x s.t. $P(X>x) = 5\%$

$\rightarrow \phi^{-1}(0.95) \approx 1.645$ is our z value

$\rightarrow 1.645 = \frac{x-\mu_X}{\sigma} \rightarrow x \approx 124.675$

(d) If 25 people are randomly selected, what is the probability that their average IQ exceeds 105? Is this probability smaller than the probability of one person exceeding 105 IQ? Why? Why not?

Reminder: For independent random variables, $Var(aX+bY) = a^2\,Var(X)+b^2\,Var(Y)$. Use this property to find the new standard deviation of the new distribution.

Let Y be the RV of 25 peoples IQ average

Know: $\mu_Y = \mu_X = 100$

Want: $Var(Y) = Var\left(\frac{X_1+X_2+\ldots+X_{25}}{25}\right) = \frac{1}{25^2} \cdot \sum_{n=1}^{25} Var(X_n) = \frac{1}{25} \cdot Var(X) = \frac{15^2}{25}$

$\rightarrow Var(Y) = \frac{15^2}{5^2} \rightarrow \sigma_Y = \sqrt{Var(Y)} = 3$

$P(Y>105) = 1-P(Y<105) = 1-\left(\phi\left(\frac{105-\mu_Y}{\sigma_Y}\right)\right) = 1-\phi\left(\frac{5}{3}\right) \approx 0.0475$

Yes, $0.0475 < 0.3707$. The probability is smaller because the sample mean has less variance than individual observations by the Central Limit Theorem.

Question 2. Adult Male Heights

The average height of adult males in Turkey is 175 cm with standard deviation 7 cm. Suppose that the male population of Turkey is 42 million people.

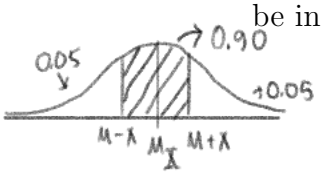
(Let X denote RV of an individual male height)

(a) Approximately how many adult males are taller than 180 cm?

Want: $42{,}000{,}000 \cdot P(X > 180) = 42m \cdot (1 - P(X < 180)) = 42m \cdot \left(1 - \phi\left(\frac{180 - \mu_X}{7}\right)\right)$

$= 42m \cdot \left(1 - \phi\left(\frac{5}{7}\right)\right) \approx 10$ million adult males.

(b) What height range contains the middle 90% of adult males? (The answer should be in the format of [μ-x, μ+x], μ being the mean)



By geometry, we need x such that $\phi\left(\frac{x - \mu_X}{\sigma}\right) = 0.95$

$\frac{x - \mu_X}{\sigma} = \phi^{-1}(0.95) \approx 1.645 \rightarrow x \approx 11.515$

So the interval is: $[163.48, 186.52]$.

(c) Ayşe is looking for a person to marry but unfortunately, only her father can choose the candidates. Also Ayşe is a picky girl, she doesn't want her future husband to be below 180 cm. Her father picks 3 Turkish male candidates. What's the probability that at least 2 of these candidates fit Ayşe's criteria so that Ayşe will at least be making a choice between at least 2 people?

$P(X > 180) = 1 - \phi\left(\frac{5}{7}\right) \approx 0.2389$

By binomial dist.:

$$\sum_{n=2}^{3} \binom{3}{n} \cdot 0.2389^{n} \cdot (1 - 0.2389)^{3-n} \approx 0.1439$$

(d) In a given region, adult males arrive at a medical screening center according to a Poisson process with rate lambda = 4 per hour. The time between arrivals follows an exponential distribution. What is the probability that the time until the next tall male (height > 180 cm) arrives exceeds 30 minutes?

$\lambda = 4 \rightarrow \lambda_{tall} = 4 \cdot P(X > 180) = 4 \cdot 0.2389 = 0.956$ tall male per hour

$P(T > 0.5) = 1 - P(T < 0.5) = 1 - (1 - e^{-\lambda \cdot 0.5}) = e^{-\lambda \cdot 0.5} = e^{-0.478} \approx 0.620$

Question 3. Intern Disaster (Mixed Random Variables)

A tech company models the weekly damage X (in thousands of TL) caused by their new intern as follows:

- With probability 0.7, the intern just makes coffee and stays quiet, so $X = 0$
- With probability 0.3, the intern touches the production server. The resulting damage follows an exponential distribution with mean 5 (thousand TL)

(a) Write the CDF $F_X(x)$ for all real values of x.

$P(X=0)=0.7$, and with probability $0.3 \rightarrow X \sim$ Exp dist with $\lambda = \frac{1}{5}$

For $x<0 \rightarrow F_X(x) = P(X<x) = 0$, for $x \geq 0 \rightarrow F_X(x) = 0.7 + 0.3 \cdot P(X<x) = 0.7 + 0.3 \cdot (1 - e^{-x/5})$

$= 1 - 0.3e^{-x/5}$

$$\rightarrow F_X(x) = \begin{cases} 0, & x<0 \\ 1-0.3e^{-x/5}, & x \geq 0 \end{cases}$$

(b) Calculate the weekly expected value (in TL) of the financial loss of the company resulted by this intern.

$E(X) = P(X=0) \cdot 0 + P(X>0) \cdot E(X \mid \text{Chaos}) = 0 + 0.3 \cdot \mu_{X \mid \text{chaos}}$

$= 0.3 \cdot 5 = 1.5$

$\rightarrow E(X) = 1{,}500$ TL

(c) Calculate $\mathrm{Var}(X)$.

$Var(X) = E(X^2) - E^2(X) \rightarrow E(X^2) = 0.7 \cdot 0^2 + 0.3 \cdot E(Y^2) = 0.3 \cdot \int_0^\infty x^2 \cdot \lambda \cdot e^{-\lambda \cdot x} dx$ (integration by parts)

$\rightarrow E(X^2)$ turns out to be 15

$Var(X) = E(X^2) - E^2(X) = 15 - 1.5^2 = 12.75$

(d) Find the probability that the intern causes a financial loss bigger than 3,000 TL, or $P(X > 3)$.

$P(X>3) = \underbrace{P(X>3 \mid \text{coffee}) \cdot P(\text{coffee})}_{0} + P(X>3 \mid \text{chaos}) \cdot P(\text{chaos})$

$= P(X>3 \mid \text{chaos}) = 1 - P(X<3 \mid \text{chaos}) = 1 - (1 - e^{-\lambda \cdot x}) \Big|_{x=3} = e^{-\lambda \cdot 3} = e^{-3/5}$

$\rightarrow P(X>3) = P(X>3 \mid \text{chaos}) \cdot P(\text{chaos}) = e^{-3/5} \cdot 0.3 \approx 0.1646$

Question 4. Sampling Distribution

The time computer science students spend on Stack Overflow, a code problem solving website, before solving a coding problem, has a mean $\mu = 50$ minutes and standard deviation $\sigma = 12$ minutes.

A research team randomly selects $n = 36$ CS students to track their Stack Overflow usage during a homework assignment.

(a) Find $E[\bar{X}]$ and $\mathrm{SD}(\bar{X})$.

$E(\bar{X}) = \mu_{\bar{X}} = \mu_X = 50$ minutes by the central limit theorem.

$SD(\bar{X}) = \sqrt{Var(\bar{X})} = \sqrt{\frac{Var(X)}{n}} = \frac{\sigma_X}{\sqrt{n}}\Big|_{n=36, \sigma_X = 12} = 2$ minutes

(b) Using the Central Limit Theorem, find $P(\bar{X} < 48)$, the probability that the average browsing time is below 48 minutes.

$P(X < 48) \rightarrow z = \frac{48 - \mu_X}{\sigma_{new}} = \frac{-2}{2} = -1 \rightarrow P(X < 48) = P(Z < -1) = 1 - \phi(1)$

≈ 0.1587

(c) Is $\bar{X}$ an unbiased estimator of μ? Why?

Yes, $\bar{X}$ is an unbiased estimator of μ because $E(\bar{X}) = \mu$.

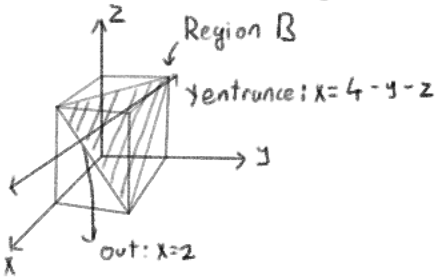
Question 5. Conditional PDF with Three Random Variables

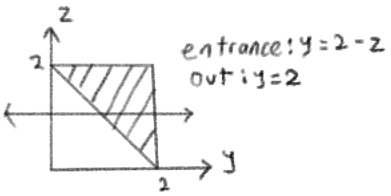
X, Y, and Z are independent random variables with joint PDF

$$f_{X,Y,Z}(x,y,z) = \begin{cases} \frac{1}{8} & 0 \le x \le 2,\ 0 \le y \le 2,\ 0 \le z \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the conditional PDF of X, Y, and Z given the event $B = \{X + Y + Z \ge 4\}$.

Want: $\iiint_{\text{over region } B} f_{X,Y,Z}(x,y,z)\,dx\,dy\,dz$. Region B is the region above the plane $x+y+z=4$.





$$\frac{1}{8}\cdot\int_0^2\int_{2-z}^2\int_{4-y-z}^2 dx\,dy\,dz = \frac{1}{8}\cdot\int_0^2\int_{2-z}^2 y+z-2\,dy\,dz = \frac{1}{8}\cdot\int_0^2 \frac{z^2}{2}\,dz = \frac{1}{8}\cdot\frac{z^3}{6}\Big|_0^2 = \frac{1}{6}.$$

$$\rightarrow f_{X,Y,Z|B}(x,y,z) = \frac{f_{X,Y,Z}(x,y,z)}{P(B)} = \frac{1/8}{1/6} = \frac{3}{4}.$$

Question 6. Student Performance Analysis

Let X = the number of hours studied and Y = exam grade be discrete random variables with the following joint PMF:

$Y \backslash X$	1	2	3
4	0.02	0.08	a
3	0.05	0.12	0.15
2	0.08	b	0.06
1	0.10	0.05	0.02

(a) Given that $P(X = 2) = 0.30$, find the values of a and b.

Using $P(X=2)=0.3$: $0.08+0.12+b+0.05=0.3 \rightarrow b=0.05$

Using $\sum_{\text{over all } (x,y)} P(X=x, Y=y)=1$ therefore sum of all prob. $=1 \rightarrow 0.78+a=1$

$\rightarrow a=0.22$

(b) Calculate $E[Y \mid X = 3]$ and $\mathrm{Var}(Y \mid X = 3)$.

$P(X=3)=0.22+0.15+0.06+0.02=0.45$

$E(Y \mid X=3)=(4\cdot 0.22+3\cdot 0.15+2\cdot 0.06+1\cdot 0.02)/0.45 \approx 3.2667$

$Var(Y \mid X=3)=\underbrace{E(Y^2 \mid X=3)}_{\sum_{\text{over all } y} y^2 \cdot P(Y \mid X=3)=\frac{513}{45} \text{ by calculation}} - E^2(Y \mid X=3) \approx 0.729$

(c) Calculate $\mathrm{Cov}(X, Y)$ and the correlation coefficient $\rho_{X,Y}$. Based on your results, does studying more hours tend to lead to higher grades? Interpret your findings.

$E(X)=2.20$, $E(Y)=2.79$, $E(XY)=6.56$

$Cov(X,Y)=E(XY)-E(X)E(Y) \approx 0.422$

$Var(X)=0.66$, $Var(Y)=1.146$

$\rho_{X,Y}=\frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} \approx 0.485 \rightarrow$ The result is positive, therefore there is a positive correlation between study time and test scores.
The correlation is neither too weak nor too strong.

Rem: $\rho=0$ is no correlation, $\rho \approx 1$ strong positive correlation, $\rho \approx -1$ strong negative correlation

Question 7. Bias of Point Estimators

Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with mean μ and variance σ^2.

Consider the following three estimators for μ:

$$\hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\hat{\mu}_2 = \frac{1}{n+1}\sum_{i=1}^{n} X_i$$

$$\hat{\mu}_3 = \frac{X_1 + X_n}{2}$$

(a) For each estimator, compute $E[\hat{\mu}_i]$ and determine whether it is biased or unbiased.

$E(\hat{\mu}_1) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\cdot\sum_{i=1}^{n}\mu = \frac{1}{n}\cdot n\cdot\mu = \mu \rightarrow \hat{\mu}_1$ is unbiased.

$E(\hat{\mu}_2) = E\left(\frac{1}{n+1}\sum_{i=1}^{n} X_i\right) = \frac{1}{n+1}\sum_{i=1}^{n} E(X_i) = \frac{n}{n+1}\mu \neq \mu$ therefore $\hat{\mu}_2$ is biased.

$E(\hat{\mu}_3) = E\left(\frac{X_1+X_n}{2}\right) = \frac{1}{2}\left(E(X_1)+E(X_n)\right) = \frac{1}{2}\cdot(\mu+\mu) = \mu$ therefore $\hat{\mu}_3$ is unbiased.

(b) For any biased estimator(s), calculate the bias: $\text{Bias}(\hat{\mu}_i) = E[\hat{\mu}_i] - \mu$.

For $\hat{\mu}_2$, $\text{Bias} = E(\hat{\mu}_2) - \mu = \frac{n}{n+1}\mu - \mu = \mu\left(\frac{n}{n+1} - 1\right)$

$= -\frac{\mu}{n+1}$.

(c) Which estimator would you prefer and why? (Consider both bias and intuition)

Prefer $\hat{\mu}_1$ estimator because:

1) It is unbiased.
2) It uses all data, n observations.
3) Amongst the 2 unbiased estimators, it has the smallest variance.

Question 8. Maximum Likelihood Estimator

A researcher is testing 5 large language models (LLMs). Each model either always answers correctly or always fails. The number of well-trained models, θ, is unknown and could be 0, 1, 2, 3, 4, or 5.

You randomly pick one of the 5 models, send it a prompt, record the result, and repeat this process 4 times (with replacement, so the same model can be picked again).

$$X_i = \begin{cases} 1 & \text{if the model answers the } i\text{th prompt correctly} \\ 0 & \text{if the model fails the } i\text{th prompt} \end{cases}$$

You observe the sequence: $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 1$ (3 correct, 1 incorrect).

(a) For each possible value of $\theta \in \{0, 1, 2, 3, 4, 5\}$, find the probability of the observed sample.

let $p = P(\text{correct}) = P(\text{well-trained LLM}) = \frac{\theta}{5}$, $1-p = 1-\frac{\theta}{5}$

$L(\theta) = \frac{\theta}{5} \cdot \left(1-\frac{\theta}{5}\right) \cdot \frac{\theta}{5} \cdot \frac{\theta}{5} = \frac{\theta^3 \cdot (5-\theta)}{625}$

$L(\theta=0) = \frac{0^3 \cdot 5}{625} = 0$, $L(\theta=1) = \frac{1^3 \cdot 4}{625} \approx 0.0064$, $L(\theta=2) = \frac{2^3 \cdot 3}{625} \approx 0.0384$

$L(\theta=3) = \frac{3^3 \cdot 2}{625} \approx 0.0864$, $L(\theta=4) = \frac{4^3 \cdot 1}{625} \approx 0.1024$, $L(\theta=5) = \frac{5^3 \cdot 0}{625} = 0$.

(b) For which value of θ is the probability of the observed sample the largest? This is your $\hat{\theta}_{MLE}$.

Max $L(\theta)$ value observed for $\theta \in \{0,1,2,3,4,5\}$ happens when $\underline{\theta = 4}$.

This corresponds to $L(4) = 0.1024$.

Question 9. Confidence Interval

A crazy scientist is measuring the training time of 150 neural network models. The mean training time was 42 minutes, with a standard deviation of 8 minutes.

(a) Find the 95% confidence interval for the mean training time of the neural network models.

Notice: S.S. $150 \geq 30$, therefore we can use the z values.

For $\alpha = 0.05$, $z_{\alpha/2} = z_{0.025} = 1.96 \rightarrow P(-1.96 \leq Z \leq 1.96) = 0.95$

$L.L. = \bar{x} - z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 42 - 1.28 = 40.72$, $UL = \bar{x} + z \cdot \frac{\sigma}{\sqrt{n}} = 43.28$

The 95% confidence interval is $(40.72, 43.28)$.

(b) Find the 99% confidence interval for the mean training time. How is it different than the answer of part (a)?

$1 - \alpha = 0.99 \rightarrow \alpha = 0.01$, $\alpha/2 = 0.005$. We need a z such that $P(Z \leq z) = 0.995$

$\phi^{-1}(0.995) = z \approx 2.58$

$L.L. = 42 - 2.58 \cdot \left(\frac{8}{\sqrt{150}}\right) \approx 40.31$, $UL = 42 + 2.58 \cdot \left(\frac{8}{\sqrt{150}}\right) \approx 43.69$

(c) If the crazy scientist wants the 95% confidence interval to have a margin of error of at most 1 minute, what is the minimum sample size required?

95% confidence interval $\rightarrow z_{0.025} = 1.96$

Margin of error $= z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$, we want $z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq 1$.

$$\frac{1.96 \cdot 8}{\sqrt{n}} = 1 \rightarrow n = (1.96 \cdot 8)^2 \approx 245.8624 \rightarrow \underbrace{\lceil 245.8624 \rceil}_{\text{Ceiling function because we want the minimum sample size.}} = \underline{246}$$

Question 10. Confidence Interval

An electrical engineer tests 15 resistors from a production batch. The sample mean resistance was 98 Ω, with a sample standard deviation of 3 Ω. Assume the resistances are normally distributed.

(a) Find the 95% confidence interval for the true mean resistance of the resistors (two-sided).

For 95% CL, $\alpha = 1 - 0.95 = 0.05$, $t_{0.025,14} = 2.145$ (by the t table) (DOF is 15-1=14)

$L.L. = \bar{x} - t_{0.025,14} \cdot \frac{\sigma}{\sqrt{n}} = 98 - 2.145 \cdot \left(\frac{3}{\sqrt{15}}\right) \approx 96.34\,\Omega$

$U.L. = \bar{x} + t \cdot \frac{\sigma}{\sqrt{n}} = 99.66\,\Omega$

Interval: (96.34, 99.66)

(b) Why do we use the t-distribution instead of the standard normal distribution in this problem?

We used the t-distribution because the sample size is smaller than 30.

(c) If the sample size were increased to 200 resistors, would you still use the t-distribution? Explain.

We can technically still use the t distribution but since 200 is larger than 30, with degree of freedom 199, it is nearly identical to the standard normal distribution, so z can also be used. Also, using z is computationally easier.

Good luck to all of you in the final exam

Batu Koray Masak